

IELTS Speaking Practice

AI IELTS Speaking Assistant — User Guide

v0.4 — August 4, 2026

Full IELTS exam simulation / AI scoring / Live2D characters / Voice interaction

Contents

1 Quick Start

1.1 Starting the Application

- **Windows users:** Double-click `start.bat` in the project root. The script will install dependencies, start both backend and frontend, and open your browser.
Visit: `http://localhost:5173`
- **Other systems:** See README.md for Docker or manual start instructions.

1.2 Configuring API Key

You need an LLM API key on first use:

1. Open the page, click **Settings** at the bottom
2. Select your model provider from the dropdown
3. Enter your API Key
4. Click **Save Configuration**

Supported providers:

Provider	Description
DeepSeek V4 Pro/Flash	Recommended, best value
OpenAI	GPT-4o-mini etc.
Groq	LPU inference, very fast
OpenRouter	Unified API gateway, many models
Ollama	Local, no paid key needed

1.3 First-time Tutorial

On your first visit, a tutorial modal will appear explaining the four key steps. Click "Got it!" to dismiss. It will not appear again. You can always revisit this guide from Settings.

2 Exam Mode

Exam mode simulates the full IELTS Speaking test.

2.1 Exam Flow

1. **AI examiner introduction** — An AI examiner introduces the test rules.
2. **Part 1: Introduction & Interview (4–5 min)** — The examiner asks 3 questions around one topic (randomly chosen from 43 topics including Hometown, Work/Study, Technology, etc.).
3. **Part 2: Individual Long Turn** — A Cue Card appears with a topic and 4 prompt points. You have 60 seconds to prepare, then speak for 1–2 minutes. The examiner may ask 1–2 follow-up questions.
4. **Part 3: Two-way Discussion** — The examiner asks 2–3 deeper discussion questions related to the Part 2 topic.
5. **Exam ends** — The examiner announces the end; the system navigates to the scoring report automatically.

2.2 Controls

- Click the **Microphone button** at the bottom to start recording
- Recording auto-stops when the timer expires; you can also stop manually
- A Live2D virtual examiner character (Haru) appears on screen, with expressions and gestures

2.3 Timer Reference

Stage	Default Time
Part 1 each question	45 seconds
Part 2 preparation	60 seconds
Part 2 speaking	120 seconds
Part 2 follow-up	60 seconds
Part 3 each question	60 seconds

3 Scoring Report

After the exam, a full scoring report is generated:

3.1 Scoring Criteria

Based on the official IELTS four criteria (0–9, 0.5 increments):

- **Fluency and Coherence** — Speech rate, pauses, logical flow
- **Lexical Resource** — Vocabulary range, precision, paraphrasing
- **Grammatical Range and Accuracy** — Sentence complexity, error frequency
- **Pronunciation** — Clarity, stress, intonation (estimated from text quality)

3.2 Overall Band Score

Average of the four criteria, rounded to the nearest 0.5.

3.3 Suggestions

The report includes 3 personalised suggestions based on your performance.

Note: Pronunciation scoring is an indirect estimate from the transcribed text. For more accurate pronunciation feedback, enable Whisper local ASR in Settings.

4 Free Chat Mode

Free Chat mode has no scoring and no time limits. It is designed for casual English practice.

- The AI greets you in English and begins a conversation
- Click the microphone to speak; the AI replies in voice
- You can change topics freely
- Chat history is not saved (cleared on refresh)
- Live2D character is Mao (casual style), different from Haru (formal, exam mode)

5 Voice Input

5.1 Browser ASR (Default)

Uses Chrome/Edge built-in Web Speech API. No extra configuration needed. Low latency, works out of the box. However, accuracy depends on the browser, and continuous recording may stop after ~60 seconds.

5.2 Whisper Local ASR

Uses faster-whisper for local speech recognition. Higher accuracy and supports longer recordings.

How to enable:

1. Go to Settings → Whisper ASR
2. Toggle **Enabled**
3. Download a model (recommended: **small**, ~460 MB)
4. Select your language
5. Save configuration

Whisper runs on CPU. Supported models range from tiny (75 MB) to large-v3 (2.9 GB) – 9 models total.

6 Appearance and Language

6.1 Live2D Characters

Two built-in Live2D virtual characters:

- **Haru (Exam mode)** — Formal style, nods, smiles, gestures
- **Mao (Free Chat mode)** — Casual, expressive

6.2 Eye Tracking

In Settings, you can set the character's gaze:

- **Follow Mouse** — The character's eyes follow your mouse
- **Look Forward** — The character looks straight ahead

6.3 UI Language and Training Language

- **UI Language:** Switch display language (English / Chinese)
- **Training Language:** Controls speech recognition target language and TTS voice. Supports English, Chinese, Japanese, Korean, French, German, Spanish (7 languages)

7 Troubleshooting

7.1 AI doesn't respond?

Check:

1. API Key is correctly entered in Settings
2. API account has sufficient credits
3. Network connection is working (some services may require a VPN)

7.2 Voice recognition not working?

1. Ensure your browser has microphone permission
2. Use Chrome or Edge (Firefox/Safari do not support Web Speech API)
3. If browser ASR frequently disconnects, try enabling Whisper local ASR

7.3 Port already in use?

The application uses port 8000 (backend) and 5173 (frontend). Close any programs using these ports and restart.

7.4 Browser didn't open automatically?

Manually navigate to `http://localhost:5173` in your browser.

7.5 How to reset the tutorial?

Open your browser's developer console (F12) and run:

```
localStorage.removeItem("tutorialSeen")
```

Then refresh the page.

7.6 How to add new exam types or languages?

See the “Adding a new exam” and “Adding a new language” sections in README.md.

8 Technical Information

Component	Technology
Frontend	React 19 / Vite 8 / TypeScript / PixiJS 6
Backend	Python FastAPI / Uvicorn
LLM	OpenAI SDK compatible (DeepSeek / OpenAI / Groq / OpenRouter / Ollama)
ASR	Web Speech API / faster-whisper
TTS	Web Speech API (Edge natural voices)
Live2D	pixi-live2d-display + Cubism 4 SDK

— *End of User Guide* —

GitHub: <https://github.com/Making-acid/ielts-speaking-ai>